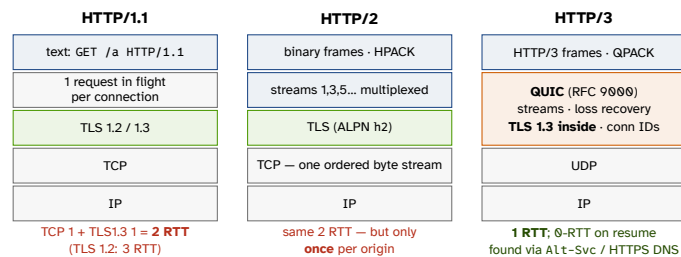
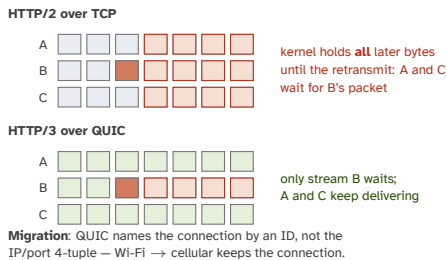


In one line: Same **semantics** (methods, status, headers — RFC 9110) on three **wire formats**: 1.1 = text, one request at a time per TCP connection; 2 = binary frames, many **streams multiplexed** on one TCP connection (TCP head-of-line blocking remains); 3 = the same over **QUIC/UDP** with TLS 1.3 built in — per-stream loss recovery, fast setup, **connection migration**.

The stacks



One lost packet



How it works

- 1.1** (RFC 9112): connections are **persistent by default** (Connection: close ends one) — reuse skips TCP + TLS setup. One response at a time: **pipelining** had to answer in order (one slow response blocks the rest) and proxies broke it — off everywhere. Workaround: ~6 connections per host.
- 2** (RFC 9113): frames (HEADERS DATA SETTINGS WINDOW_UPDATE RST_STREAM PING GOAWAY); client streams have odd ids; flow control per stream *and* connection. RST_STREAM cancels one request without killing the connection. **HPACK**: static + dynamic table + Huffman — a repeated cookie costs a byte or two. Server push: dead (Chrome removed it, 2022 (*unverified*)) — use 103 Early Hints.
- 3** (RFC 9114): streams are independent in QUIC, so loss stalls only its stream. **QPACK** replaces HPACK (HPACK assumes in-order delivery). UDP blocked → the client falls back to TCP.

Caching (RFC 9111) — freshness, then validation

max-age=60	fresh for 60 s — served with no request
no-cache	may store , must revalidate before every use
no-store	never write it anywhere (secrets, PII)
private / public	only the end client / shared caches (CDN) too
s-maxage	freshness for shared caches only
must-revalidate	no stale reuse after expiry
stale-while-revalidate	serve stale now, refresh in background
immutable	never revalidate (hashed asset URLs)

Validation: response ETag: "v7" (opaque; W/ = weak) or Last-Modified → request If-None-Match / If-Modified-Since → **304** (no body, reuse cache) or 200.
Vary: **Accept-Encoding** adds those request headers to the cache key; Vary: * makes it uncacheable.

Example — which protocol did I get?

```
final class Probe: NSObject, URLSessionTaskDelegate {
    func URLSession(_ s: URLSession, task: URLSessionTask,
        didFinishCollecting m: URLSessionTaskMetrics) {
        for t in m.transactionMetrics {
            print(String(describing: t.networkProtocolName), // h2, h3
                t.isReusedConnection, // keep-alive hit
                t.resourceFetchType) // .localCache?
        }
    }
}
var req = URLRequest(url: url)
req.assumesHTTP3Capable = true // race QUIC (iOS 14.5+)
```

What URLSession does for you

- Negotiates **h2** via ALPN and **h3** once the server advertises it — no code. Pools and reuses connections *per session*.
- Sends Accept-Encoding: gzip, deflate, br and decodes transparently; *request* bodies are never compressed for you. expectedContentLength may be -1.
- URLCache obeys Cache-Control; when *it* revalidates, a 304 reaches you as 200 + the cached body (*unverified*). Responses over ~5 % of the disk cache are not stored.
- Cookies: HTTPCookieStorage (httpShouldSetCookies). WKWebView keeps its own WKHTTPCookieStore — sync them yourself.

CORS — a browser rule

The **browser** blocks JS from *reading* a cross-origin response unless the server opts in (Access-Control-Allow-Origin). “Simple” requests (GET/HEAD/POST with form types, no custom headers) go straight out; others — application/json, Authorization, PUT/DELETE — get an **OPTIONS preflight** (Access-Control-Request-Method/-Headers → Allow-Methods/-Headers/-Max-Age). Allow-Origin: * is refused with credentials. URLSession never enforces it; JS inside WKWebView does.

Cookies

Secure (HTTPS only) · HttpOnly (no JS — limits XSS theft) · SameSite=Strict/Lax/None (cross-site sending — the CSRF defence; None needs Secure; Chrome defaults to Lax) · prefix __Host- = Secure, no Domain, Path=.

Traps

- ! no-cache is not “don't cache” — that is no-store.
- ! “HTTP/2 removed head-of-line blocking” — only at the HTTP layer; TCP still stalls all streams.
- ! Setting If-None-Match yourself: now you get the raw 304 with an empty body — decode fails.
- ! Per-user response without private: a CDN can serve user A's data to user B.
- ! “CORS error in my iOS app” — impossible in URLSession; and CORS protects users' browsers, it is not API auth.
- ! 1.1-era domain sharding hurts h2: one connection is the point.

Remember

“1.1 queues, 2 multiplexes over one pipe, 3 gives every stream its own pipe.”